Shao-En Lin

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SUMMARY

M.S. in Data Science with experience in graph machine learning and recommendation systems, including two first-author publications at top AI conferences (*ICLR 2026*, *WWW 2025*). Interested in Software Engineering and AI/ML roles.

EXPERIENCE

iDSSP Lab, National Taiwan University **Jul 2023 – Jan 2026**
Research Assistant Taipei, Taiwan

- Conducted research in graph ML and recommendation systems, resulting in 2 first-author publications (*ICLR 2026*, *WWW 2025*).
- Led end-to-end research cycles including problem formulation, modeling, large-scale experimentation, and paper writing.
- Built internal CLI utilities and documentation to standardize cloud-based training and experimentation workflows.

EDUCATION

National Taiwan University **Jul 2023 – Jan 2026**
M.S. in Communication Engineering, Data Science Group

- GPA: 4.17/4.3
- Courses: Convex Optimization, Neural Networks, Financial Technology
- Thesis: *Entity-Attribute Duality for Heterogeneous Graph Modeling*
- Advisor: Prof. *Che Lin*

National Cheng Kung University **Jul 2019 – Jun 2023**
B.S. in Electrical Engineering

- GPA: 3.60/4.3
- Courses: Data Structures, Probability and Statistics, Operating Systems, Algorithms, Theory of Computation

PROJECTS

AtomHIN | *ICLR 2026* | *GitHub* **2026**
Graph ML, GNNs

- Designed a principled schema framework for heterogeneous graphs, achieving consistent performance across 8 datasets.
- Led end-to-end research and system development, releasing a Python library and benchmark datasets to support practical adoption.

BETag | *WWW 2025* | *GitHub* **2025**
LLMs, Recommendation Systems

- Developed an LLM-based item tagging system incorporating user behavior signals for recommendation pipelines.
- Achieved up to 46% (user-based) and 26% (item-based) improvements over strong tagging baselines; led a two-person collaboration from design to implementation.

SynHING | *arXiv* **2024**
Graph, Algorithms, GNNs

- Improved synthetic HIN generation by reducing merge complexity from $O(n^2)$ to $O(n)$, enabling million-scale graph generation in under 10 minutes.

SKILLS

Programming Python

ML/AI GNNs, LLMs, Recommendation Systems

Frameworks PyTorch, HuggingFace, DGL

Tools & Libraries Git, Linux, NumPy, Scikit-learn, Pandas, Pydantic

Languages Mandarin Chinese, English

CERTIFICATIONS

TOEIC 940/990 **Jun 2024**

SELECTED ACHIEVEMENTS & CONTRIBUTIONS

2024 National Taiwan University FinTech Best Project Award (First Place).

2022 Open-source contributor to PyTorch (merged GitHub PR).

2022 Ahk-Doc: VS Code language service for AutoHotkey (1,763+ downloads).

PUBLICATIONS

1. **Shao-En Lin**, Ming-Yi Hong, Miao-Chen Chiang, Chih-Yu Wang, Che Lin. (2026) *Atomic HINs: Entity-Attribute Duality for Heterogeneous Graph Modeling*. *International Conference on Learning Representations (ICLR '26)*.
2. Ming-Yi Hong, You-Chen Teng, **Shao-En Lin**, Chih-Yu Wang, Che Lin. (2026). *HINPool: A Unified Heterogeneous Graph Pooling Framework for Accurate Molecular and Protein Property Prediction*. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI '26)*.
3. **Shao-En Lin**, Brian Liu, Miao-Chen Chiang, Ming-Yi Hong, Yu-Shiang Huang, Chuan-Ju Wang, and Che Lin. (2025). *BETag: Behavior-enhanced Item Tagging with Fine-tuned Large Language Models*. In *Proceedings of the ACM on Web Conference 2025 (WWW '25)*.
4. Ming-Yi Hong, Yi-Hsiang Huang, **Shao-En Lin**, You-Chen Teng, Chih-Yu Wang, Che Lin. (2024). *SynHING: Synthetic Heterogeneous Information Network Generation for Graph Learning and Explanation*.